

ABU BAKAR YASIR

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SUMMARY

Full-Stack and AI Engineer with 1.5 years of professional experience building production software. Sole developer of an AI platform that monitors UK government and media activity and writes intelligence briefings automatically, replacing research work that public affairs teams did by hand. Works across backend architecture, AI and retrieval systems, database design, and frontend.

EXPERIENCE

Full-Stack & AI Engineer — [Spiral Lab](#)

April 2025 – Present

Delivered 2 production client platforms, deployed on Railway and Vercel with automated CI.

[Magnai — AI Platform for UK Public Affairs](#) | Django REST Framework, PostgreSQL, Anthropic Claude, OpenAI, Django-Q2, Redis
Monitors UK government and media activity and writes intelligence briefings automatically for public affairs teams.

- Sole developer. Designed an AI orchestrator coordinating 9 sub-agents and 137 tools, giving analysts one system for UK, Scottish, Welsh, and Northern Irish political data that previously had to be tracked source by source.
- Integrated 15 external data sources — UK Parliament, 4 government portals, 32 regulator feeds, TheyWorkForYou, Opoint, NewsAPI, and 6 Twitter/X endpoints — so each briefing draws on parliamentary, regulatory, and media activity in a single pass.
- Built background processing on Django-Q2 and Redis running 884 scheduled jobs daily, keeping data current around the clock unattended; race-condition handling and 10-minute uptime checks let the pipeline recover from failures without intervention.
- Developed a stakeholder-ranking engine using a 22-factor weighted algorithm, surfacing which politicians and officials matter most for a given client issue instead of analysts building those lists by hand.
- Implemented multi-tenant access control with 4 permission roles, keeping each client organisation's data isolated. Wrote 1,459 automated tests across 65 files, enabling continuous deployment without manual regression checks.

[Lynxflow-Health — Healthcare Management Platform](#) | Next.js, Supabase (PostgreSQL), REST APIs

Lets clinics manage patients, staff permissions, and connected medical devices in one system.

- Owned frontend development and database engineering within a cross-functional team.
- Designed the database schema and a 6-tier permission hierarchy (Super-Super Admin, Super Admin, Facility Admin, Doctor, Clinician, Patient), enforced through row-level security so each facility sees only its own patient records and staff access matches clinical responsibility.
- Built role-aware Next.js interfaces so each role lands on the tools relevant to it rather than a shared dashboard, and modelled patient records for HIPAA-aligned handling.

PROJECTS

[neuroMedica — Medical AI for Clinical Decision Support](#) | FastAPI, Next.js, Qdrant, PyTorch, TensorFlow, Neo4j, Docker

Answers clinical questions from medical literature and flags conditions in ECG and X-ray scans.

- Trained a DenseNet-121 chest X-ray classifier on NIH ChestX-ray14 reaching 0.846 mean AUC across 14 conditions on 8,557 validation images, in line with published baselines for this dataset.
- Trained an ECG classifier on PTB-XL (21,430 records): 0.926 macro AUC across 5 diagnostic categories.
- Built a hybrid retrieval pipeline combining semantic (PubMedBERT/Qdrant) and keyword (BM25) search via Reciprocal Rank Fusion over 8,399 chunks from 2,651 documents, so answers are grounded in cited medical literature rather than model memory.
- Added a self-evaluation loop that grades and discards low-relevance context before generation, reducing the risk of confident but unsupported clinical answers.

EDUCATION

BS Computer Engineering — COMSATS University Islamabad, Lahore

Graduated June 2026

CGPA 3.29 / 4.0

SKILLS

Languages: Python, JavaScript, TypeScript, SQL, C++

AI / ML: RAG, Multi-Agent Orchestration, LLM Integration (OpenAI, Anthropic Claude, Groq, Gemini), Embeddings, Prompt Engineering, PyTorch, TensorFlow

Backend: Django, Django REST Framework, FastAPI, Node.js, REST APIs, WebSockets, Redis

Frontend: Next.js, React.js, TypeScript, Tailwind CSS

Databases: PostgreSQL, Supabase, MongoDB, MySQL, Qdrant, ChromaDB, Neo4j

Practices: Schema design, Row-Level Security, RBAC, Automated testing, CI/CD, Docker, Git